

CO-PO-PSO Mapping & Attainment

CS401 — Computer Architecture and Organization

Auqib Hamid Lone

Department of Computer Science & Engineering, GCET Ganderbal

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Review required before filing. *This document is a faculty-review **draft**, not an authoritative NBA/AICTE filing. The CO-PO-PSO correlation matrix (Section 3) reflects each Course Outcome’s stated content, not verified pedagogical practice, and every numeric attainment value (Section 4) is marked [FILL] — no student assessment score data exists in this repository. The instructor must review, correct, and sign off on both before submission.*

1 Course Outcomes

Numbered here for the first time from `syllabi/CS401.md`’s Learning Objectives, in delivery order. Bloom’s-taxonomy levels are a first-pass draft inferred from each objective’s stated verb; objectives whose verbs span two levels are flagged rather than silently collapsed to one.

2 Assessment-Tool Mapping

Cross-referenced against `syllabi/CS401.md`’s own Assessment table (Class Tests 1–3, Assignments 1–4, End-semester exam, the Nand2Tetris practical labs) — no new assessment data is invented here, only cross-referenced.

CO	Statement	Bloom level		Weeks
CO1	Compute and compare system performance using the basic performance equation, clock rate, and Amdahl's Law	Apply (flag: "com- pare")		1
CO2	Represent and manipulate numbers in fixed-point and IEEE754 floating-point formats, and perform arithmetic/logical operations in hardware	Apply		2-3
CO3	Describe instruction sets, addressing modes, and CPU register/bus organization, and trace an instruction cycle	Understand/Apply (flag)		4-5
CO4	Design and contrast hardwired and microprogrammed control units for a given instruction set	Create (flag: "con- trast")		6-7
CO5	Analyze I/O techniques (interrupt, polling, DMA, IOP) and select the appropriate technique for a given latency/throughput requirement	Analyze		8
CO6	Evaluate memory hierarchy performance, including cache mapping/replacement policy choice and virtual memory (paging/segmentation) design	Evaluate (flag: "de- sign")		9-10
CO7	Compare Flynn's taxonomy classes and identify/resolve pipeline hazards in a pipelined instruction stream	Analyze		11
CO8	Contrast RISC and CISC architectural philosophies and justify a design choice for a given workload	Evaluate (flag: "con- trast")		12

Table 1: Course Outcomes, copied verbatim from `syllabi/CS401.md`.

CO	Assessed in
CO1	Class Test 1, Assignment 1, End-semester exam
CO2	Class Test 1, End-semester exam, Lab 1-2 (Nand2Tetris)
CO3	Class Test 2, Assignment 2, End-semester exam, Lab 4 (Nand2Tetris)
CO4	Class Test 2, End-semester exam
CO5	Class Test 3, Assignment 3, End-semester exam
CO6	Class Test 3, End-semester exam
CO7	Assignment 4, End-semester exam
CO8	End-semester exam

Table 2: Every CO maps to at least one existing assessment tool – no unassessed CO.

3 CO-PO-PSO Correlation Matrix

Review required before filing. *Draft correlation matrix — requires instructor review and sign-off before submission. Correlation strengths (1 = slight, 2 = moderate, 3 = substantial, blank = no correlation) reflect the CO's stated content, not verified pedagogical practice. PSO columns are [FILL]: Program-Specific Outcomes are department-defined and not yet standardized anywhere in this repository.*

NBA mandates exactly these 12 Program Outcomes for every accredited UG engineering program: PO1 Engineering knowledge, PO2 Problem analysis, PO3 Design/development of solutions, PO4 Conduct investigations of complex problems, PO5 Modern tool usage, PO6 The engineer and society, PO7 Environment and sustainability, PO8 Ethics, PO9 Individual and team work, PO10 Communication, PO11 Project management and finance, PO12 Life-long learning.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3										1	[FILL]	[FILL]
CO2	3	2			2							1	[FILL]	[FILL]
CO3	3	2			2								[FILL]	[FILL]
CO4	3	2	3		1								[FILL]	[FILL]
CO5	2	3	2										[FILL]	[FILL]
CO6	3	2	2										[FILL]	[FILL]
CO7	2	3											[FILL]	[FILL]
CO8	2	2	1									2	[FILL]	[FILL]

Table 3: Draft CO-PO-PSO correlation matrix (1/2/3 scale; blank = no correlation).

4 Attainment Methodology

Standard NBA formulas, with every numeric slot left as [FILL] — no student assessment score data exists in this repository (no /grade skill has been built yet). Once real per-student, per-question scores exist, mapped to the CO each question tests via Table 2, the direct-attainment percentages below can be computed and substituted.

- **Direct attainment (per CO):** percentage of students scoring at or above the target threshold (typically 60%) in the assessment(s) mapped to that CO. Attainment level: Level 3 if [FILL]% of students meet the target, Level 2 if [FILL]%, Level 1 if [FILL]% (institute-specific cutoffs, not yet fixed in this repository).
- **Indirect attainment (per CO):** from a course-exit survey (student self-assessment), normalized to the same 1–3 scale. [FILL] — no survey instrument or data exists yet.
- **Overall CO attainment:** $0.8 \times \text{direct} + 0.2 \times \text{indirect}$ (weights are institute policy; adjust if your department's weighting differs).
- **PO attainment:** for each PO, the weighted average of the overall CO attainments mapped to it, using the correlation levels in Table 3 as weights:

$$\text{PO}_j \text{ attainment} = \frac{\sum_i \text{corr}(\text{CO}_i, \text{PO}_j) \times \text{overall attainment}(\text{CO}_i)}{\sum_i \text{corr}(\text{CO}_i, \text{PO}_j)}$$

5 Faculty Sign-Off

I have reviewed the Course Outcomes (Section 1), the assessment-tool mapping (Section 2), and the CO-PO-PSO correlation matrix (Section 3) above, and confirm they accurately reflect this offering of CS401.

Name: _____

Date: _____